Day 2 of 15 | Core Unit: Apply Statistical Concepts and Analysis

Descriptive Statistics in Excel

Student Handout · TESDA Data Analytics Level III · Batch DAL3-2026

Case company: **Tindahan ni Aling Nena Mini-Mart** · 90 days of branch sales (1 Jan – 31 Mar 2026)

Learning Outcomes

By the end of Day 2 you should be able to:

1. Compute and interpret the **mean, median, and mode**, and choose the right one for the data.
2. Compute and interpret **range, variance, standard deviation, and coefficient of variation**.
3. Find **quartiles, the IQR, and the five-number summary**, and detect **outliers** with the 1.5 × IQR rule.
4. Build a **frequency distribution** and a **histogram**, and describe its **shape** (symmetric or skewed).
5. Use Excel functions **and** the **Analysis ToolPak**, and **cross-check** results (quality standard).
6. Write plain-language **insights** for a business owner.

Materials you need today

Item	File / Location
Branch sales dataset (90 days × 4 branches)	`Day02/assets/Day02_Branch_Sales.xlsx`
Activity template	`Day02/assets/Day02_Descriptive_Stats_Template.xlsx`
Demo video (with captions)	`Day02/demo/Day02_Demo_Descriptive_Statistics_in_Excel.mp4`
Activity sheets	`Day02_Student_Activities.pdf`

Part 1 — What Are Descriptive Statistics?

Descriptive statistics summarize a dataset with a few numbers and charts, so we can understand it quickly. They answer four questions:

Question	Measures
Where is the **center**?	Mean, median, mode
How **spread out** is the data?	Range, variance, standard deviation, IQR, coefficient of variation
What is the **position** of a value?	Minimum, quartiles, percentiles, maximum
What is the **shape**?	Frequency distribution, histogram, skewness

Worked example used throughout this handout: Caloocan branch daily sales (₱) for 7 days:

`9,800 11,200 10,500 12,000 10,500 3,200 13,100`

Sorted: `3,200 9,800 10,500 10,500 11,200 12,000 13,100` (n = 7)

Part 2 — Measures of Central Tendency

Mean (arithmetic average)

Mean = sum of values ÷ number of values $\bar{x} = \Sigma x \div n$

Example: 70,300 ÷ 7 = **₱10,042.86** Excel: `=AVERAGE(range)`

Median (middle value of the sorted data)

- If n is odd, the median is the middle value. If n is even, it is the average of the two middle values.
- Position of the median = (n + 1) ÷ 2 = (7 + 1) ÷ 2 = 4th value → **₱10,500** Excel: `=MEDIAN(range)`

Mode (most frequent value)

- ₱10,500 appears twice → **mode = ₱10,500** Excel: `=MODE.SNGL(range)`. Use `=MODE.MULT(range)` when there are several modes.
- The mode works best for **discrete or categorical** data, such as the number of transactions or the most popular payment method. Continuous peso amounts rarely repeat, so Excel may return **#N/A**, meaning no value repeats.

Which average should I use?

Situation	Best measure	Why
Symmetric data, no extreme values	**Mean**	Uses every value
Skewed data or outliers (e.g., the 3,200 flood day)	**Median**	Not pulled by extreme values
Categories / most common item	**Mode**	The only valid "average" for nominal data

> **See the effect of an outlier:** without the ₱3,200 flood day, the mean jumps from ₱10,042.86 to **₱11,183.33**, while the median moves only from ₱10,500 to **₱10,850**. The median is **resistant** to outliers; the mean is not.

Part 3 — Measures of Dispersion (Spread)

Two branches can have the same average and very different consistency. Spread measures how far values are from the center.

Range

Range = Maximum − Minimum = 13,100 − 3,200 = **₱9,900** Excel: `=MAX(r)-MIN(r)`
Easy to compute, but it depends on only two values, so one outlier changes it a lot.

Variance and standard deviation

1. Find each value's **deviation** from the mean $(x - \bar{x})$.
2. **Square** each deviation, so negatives don't cancel out the positives.
3. Add them up. For a **sample**, divide by **n − 1**; for a whole **population**, divide by **N**.
4. The **standard deviation** is the square root of the variance. It is back in pesos, so it is easy to interpret.

x	$x - \bar{x}$	$(x - \bar{x})^2$
3,200	−6,842.86	46,824,693.88
9,800	−242.86	58,979.59
10,500	457.14	208,979.59
10,500	457.14	208,979.59
11,200	1,157.14	1,338,979.59
12,000	1,957.14	3,830,408.16
13,100	3,057.14	9,346,122.45
Σ	**0**	**61,817,142.86**

Sample variance s^2 = 61,817,142.86 ÷ (7 − 1) = **10,302,857.14**
Sample standard deviation s = √10,302,857.14 = **₱3,209.81**

Excel	Use for
`=VAR.S(r)` / `=STDEV.S(r)`	**Sample** data (most business data, e.g., 90 days out of all possible days)
`=VAR.P(r)` / `=STDEV.P(r)`	The **entire population** is in the data

Interpretation: "Caloocan's daily sales are typically about **₱3,210 above or below** the average of ₱10,043."

Coefficient of variation (CV)

CV = (standard deviation ÷ mean) × 100% = 3,209.81 ÷ 10,042.86 × 100 = **31.96%**
Use CV to compare consistency between branches of **different sizes**. A lower CV means more stable sales.

Part 4 — Quartiles, IQR, Five-Number Summary, and Outliers

Quartiles divide sorted data into four equal parts.

Measure	Meaning	Excel	Example
Minimum	Smallest value	`=MIN(r)`	3,200

Measure	Meaning	Excel	Example
Q1	25% of values are below it	`=QUARTILE.INC(r,1)`	10,150
Median (Q2)	50% below	`=MEDIAN(r)`	10,500
Q3	75% below	`=QUARTILE.INC(r,3)`	11,600
Maximum	Largest value	`=MAX(r)`	13,100

These five values are the **five-number summary**, which is also what a **box plot** shows.

Interquartile range (IQR) = Q3 − Q1 = 11,600 − 10,150 = **1,450**. It is the spread of the middle 50%, and outliers do not affect it.

The 1.5 × IQR outlier rule

- **Lower fence** = Q1 − 1.5 × IQR = 10,150 − 2,175 = **7,975**
- **Upper fence** = Q3 + 1.5 × IQR = 11,600 + 2,175 = **13,775**
- Any value **below 7,975 or above 13,775** is an **outlier** → **₱3,200 is an outlier.**

Excel flag formula: `=IF(OR(B2<$F$1,B2>$F$2),"OUTLIER","")`

> **An outlier is not automatically an error.** Investigate it first:
> - **Data error** (typo, duplicate) → correct it.
> - **Real event** (flood, fiesta, payday, promo) → keep it, explain it, and consider using the median.
>
> Never delete data just because it is inconvenient. This is part of applying **quality standards** and **ethical work practices**.

Part 5 — Frequency Distribution, Histogram, and Shape

A **frequency distribution** groups values into classes (bins) and counts how many fall in each.

Steps

1. Find the minimum and maximum.
2. Choose about 5–12 classes of **equal width**. Example: width ₱4,000 → 0–4,000, 4,001–8,000, …
3. Count the values in each class: `=COUNTIFS(r,">="&lower, r,"<="&upper)` or `=FREQUENCY(data, bins)`.
4. Add the **relative frequency** (%) = f ÷ n × 100 and the **cumulative frequency**.

A **histogram** is a bar chart of the frequency distribution, with bars touching because the classes are continuous.
Excel 2016 and later: select the data → **Insert** → **Insert Statistic Chart** → **Histogram**. You can also build a column chart from your frequency table and set the gap width to 0–5%.

Shapes of distributions

Shape	What it looks like	Mean vs median	Skewness (`=SKEW(r)`)
Symmetric (bell-shaped)	Mirror image around the center	Mean ≈ median	≈ 0 (between −0.5 and 0.5)
Right (positively) skewed	Long tail to the right; a few very high values	Mean > median	> 0
Left (negatively) skewed	Long tail to the left; a few very low values	Mean < median	< 0

Sales data is often **right-skewed** because of promo days, paydays, and fiestas.

Part 6 — Analysis ToolPak: Descriptive Statistics in One Click

Enable it once: File → Options → **Add-ins** → at the bottom, Manage: **Excel Add-ins** → **Go…** → ✓ **Analysis ToolPak** → OK.
The **Data Analysis** button appears at the far right of the **Data** tab.

Run it: Data → **Data Analysis** → **Descriptive Statistics** → OK.

- **Input Range:** the data, including headers (e.g., `Daily_Sales!$C$1:$F$91`)
- **Grouped By:** Columns · ✓ **Labels in first row**
- **Output Range:** an empty cell, or **New Worksheet Ply**
- ✓ **Summary statistics** → OK

Output includes Mean, Standard Error, Median, Mode, Standard Deviation (sample), Sample Variance, Kurtosis, Skewness, Range, Minimum, Maximum, Sum, and Count.

> ⚠️ The ToolPak output is **static values**. It does not update when the data changes; re-run it after edits. Excel functions update automatically.

Quality check: cross-verify

A professional analyst computes key numbers **two ways** and confirms they match. Today you compute with **functions** (Activity 2.1), then with the **ToolPak** (Activity 2.2), and compare them. If they don't match, look for a wrong range, a missing row, or STDEV.P used instead of STDEV.S. *(Integrated: Apply quality standards.)*

Part 7 — From Numbers to Insights

Numbers alone don't help Aling Nena. **Translate** them into plain sentences.

Weak	Strong
"QC mean = 19,697."	"Quezon City sells about **₱19,700 a day**, the highest of the four branches."
"Caloocan CV = 21.6%."	"Caloocan's sales are the **least predictable** (CV 21.6%), so stock should have a bigger safety buffer."
"Pasig has 3 outliers."	"Pasig had **3 unusual days**. The biggest was the **7 March barangay fiesta (₱30,606)**, so plan extra stock for the next fiesta."

Insight formula: *What* (the number) + *So what* (what it means) + *Now what* (a recommendation).

Excel Function Summary

Measure	Function
Count	`=COUNT(r)`
Mean	`=AVERAGE(r)`
Median	`=MEDIAN(r)`
Mode	`=MODE.SNGL(r)` · `=MODE.MULT(r)`
Min / Max	`=MIN(r)` · `=MAX(r)`
Range	`=MAX(r)-MIN(r)`
Variance (sample / population)	`=VAR.S(r)` · `=VAR.P(r)`
Std deviation (sample / population)	`=STDEV.S(r)` · `=STDEV.P(r)`
Quartiles	`=QUARTILE.INC(r,1)` · `=QUARTILE.INC(r,3)`
Percentile	`=PERCENTILE.INC(r,0.9)`
Skewness	`=SKEW(r)`
Frequency	`=COUNTIFS(r,">="&a,r,"<="&b)` · `=FREQUENCY(data,bins)`

Glossary

Term	Definition
Mean	Sum of values divided by their count
Median	Middle value of sorted data
Mode	Most frequent value
Range	Maximum minus minimum

Term	Definition
Variance	Average squared deviation from the mean (n − 1 for samples)
Standard deviation	Square root of the variance; typical distance from the mean
Coefficient of variation	SD as a percentage of the mean; compares relative spread
Quartile	A value that divides sorted data into quarters
IQR	Q3 − Q1, the spread of the middle 50%
Outlier	A value unusually far from the rest (outside the 1.5 × IQR fences)
Frequency distribution	A table of counts per class
Histogram	A bar chart of a frequency distribution
Skewness	A measure of a distribution's asymmetry

Key Takeaways

- Report a **center and a spread** together, e.g., "₱19,700 ± ₱3,540 a day".
- With **outliers or skew**, prefer the **median and IQR**.
- **Investigate outliers**; never delete them blindly.
- **Cross-check** results (functions vs ToolPak), and turn numbers into **insights** a business owner can use.

Next session (Day 3): Probability, the normal distribution, z-scores, correlation, and simple regression.